

Smart Contract Developer Roadmap (Solidity)

Introduction

This roadmap is designed for aspiring smart contract developers who want to master Solidity and start building decentralized applications on Ethereum.

Phase 1: Solidity Basics (1-1.5 months)

- What is Blockchain, Ethereum, and EVM
- Remix IDE setup for Solidity
- Solidity Basics: Data Types, Variables, Functions, Control Flow
- Smart Contract Structure
- Mappings and Arrays
- Events and Emit
- Modifiers (e.g., onlyOwner)
- Sending & Receiving Ether
- Error Handling (require, assert, revert)

Projects: Storage contract, Counter contract, Ether wallet

Phase 2: Advanced Solidity (1 month)

- Inheritance and Abstract Contracts
- Interfaces and ERC Standards (ERC-20, ERC-721)
- Reentrancy and Gas Optimization
- delegatecall, selfdestruct
- Fallback & Receive functions
- Upgradeable Contracts

Projects: Custom Token, NFT, Voting App, Crowdfunding App, Multi-sig Wallet

Phase 3: Blockchain Tools (0.5-1 month)

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- Hardhat or Truffle (Dev Frameworks)
- Writing and Running Tests (JavaScript)
- Deploying to Ethereum Testnets (Goerli, Sepolia)
- MetaMask Integration
- RPC Providers: Alchemy, Infura
- IPFS for decentralized storage

Project: DApp to mint and display NFTs

Phase 4: Practice and Launch

- Platforms: CryptoZombies, Ethernaut, SpeedRunEthereum, Alchemy University
- Build Portfolio (GitHub, Personal Website)
- Join Web3 Communities (Discord, Twitter, GitHub)
- Apply to internships, hackathons, and junior roles

Total Timeline

- Solidity Basic: 1-1.5 months
- Advanced Solidity: 1 month
- Tools (Hardhat, MetaMask): 15-20 days
- Projects & Practice: Ongoing

Total Duration: ~3 months to become job-ready.